

BUILD WITH AI: CODE FOR COMMUNITIES · GDG INDIA

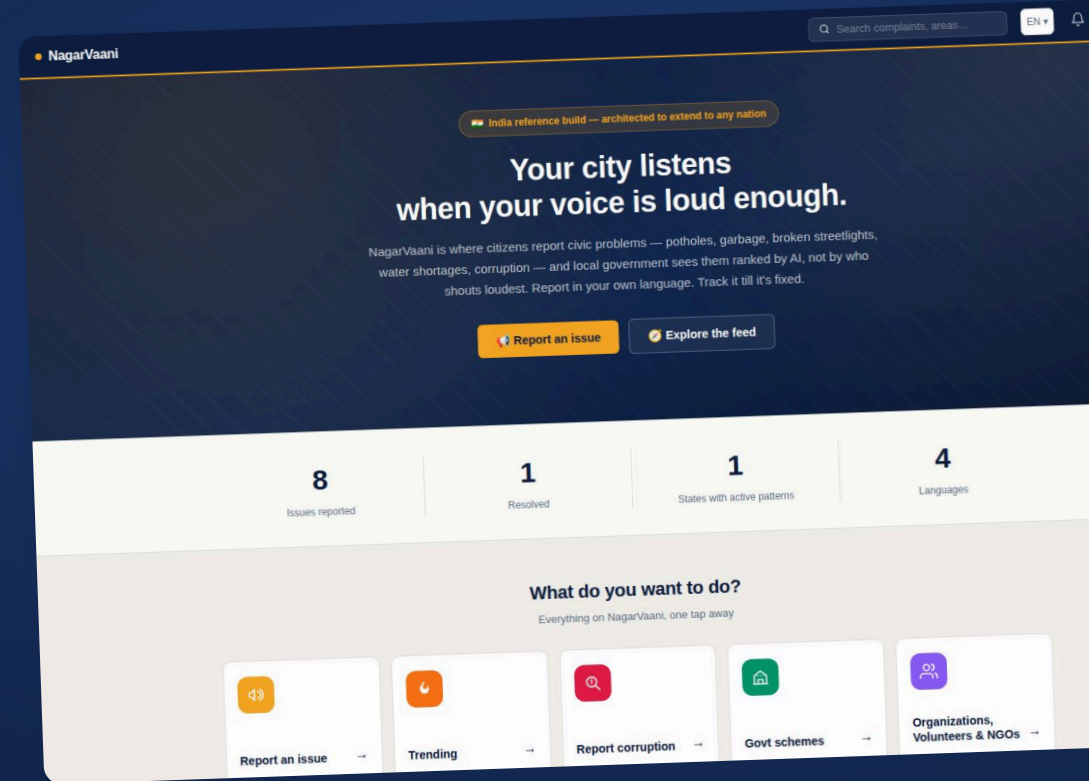
# NagarVaani नगरवाणी

## Your Voice, Your City.

A civic-intelligence layer that ranks public problems by **how urgent they really are** — not by who shouts loudest.

 <https://nagarvaani-2i2l.onrender.com>

Built by Falguni Takawale · MIT-licensed  
[github.com/FalguniTakawale/NagarVaani](https://github.com/FalguniTakawale/NagarVaani)



# Citizens speak. Governments can't hear it in one place.



## Fragmented

Requests are scattered across helplines, letters, WhatsApp groups, social media and separate portals — nobody sees the whole picture.



## The loudest wins

Ranking by votes, recency or city size rewards big cities and busy audiences — and buries villages, minority languages and problems people have stopped hoping will be fixed.



## Nothing is measured

Spending drifts away from real need, infrastructure gaps stay open, and there is no way to tell whether an initiative actually changed anything.

**The challenge: a scalable, multilingual AI platform that turns citizen requests into priorities policymakers can act on.**

# Low votes ≠ low importance.

Urgency is a property of the **problem**, not of its audience. A village with one contaminated handpump must outrank a busy road with a hundred upvotes about potholes.

PRIORITY SCORE (0–100) · FROM OUR REAL SCORING CODE, DEMO DATA

 Village handpump, discoloured water · **1 vote**

98.7

 Potholes on a busy road · **104 votes**

40.6

Votes can't outrank danger — they add a small, bounded nudge.

- 1 **Urgency ≠ audience.** Whether water is entering homes doesn't depend on who saw the post.
- 2 **Problems repeat.** One blocked drain is often one drain line across five wards — fix it once.
- 3 **Votes serve accountability,** not triage — they show attention and whether issues ever get resolved.
- 4 **Let people speak** — in their own language, by voice or chat.

# From a spoken complaint to a ranked action — in five steps.



## 1 · Intake

Web form, **voice note** or **Telegram**. Hindi, Marathi, Tamil, English — even Hindi typed in English letters.



## 2 · Filter

AI rejects political or communal framing and suggests a neutral rewording.



## 3 · Classify

AI reads category, language, location and "is someone likely to be hurt soon?" and translates to English.



## 4 · Score

**Plain arithmetic, no AI.**  
L1–L5 gives a 0–100 priority you can explain line by line.



## 5 · Act

Ranked queue, AI brief, map. Status updates reach citizens and voters; a false "resolved" can be disputed.

### AI reads. Maths ranks.

The language model never decides who gets help first — that stays transparent and auditable.

### Fails open, not closed

If the AI service is down, complaints are still accepted; nothing is lost.

### Provider-agnostic

Gemini by default, Claude in one line; Whisper via Hugging Face, Groq or OpenAI.

## THE SCORING SYSTEM

# L1 – L5: priority you can read.

**L1 Safety** — could someone be hurt in days? **90–99 band**

**L2 Season** — monsoon drains, summer water & power **×1.0–2.0**

**L3 Type** — water 28 · drainage 26 · electricity 22 ... **+10–28**

**L4 People** — log-scaled population **up to +15**

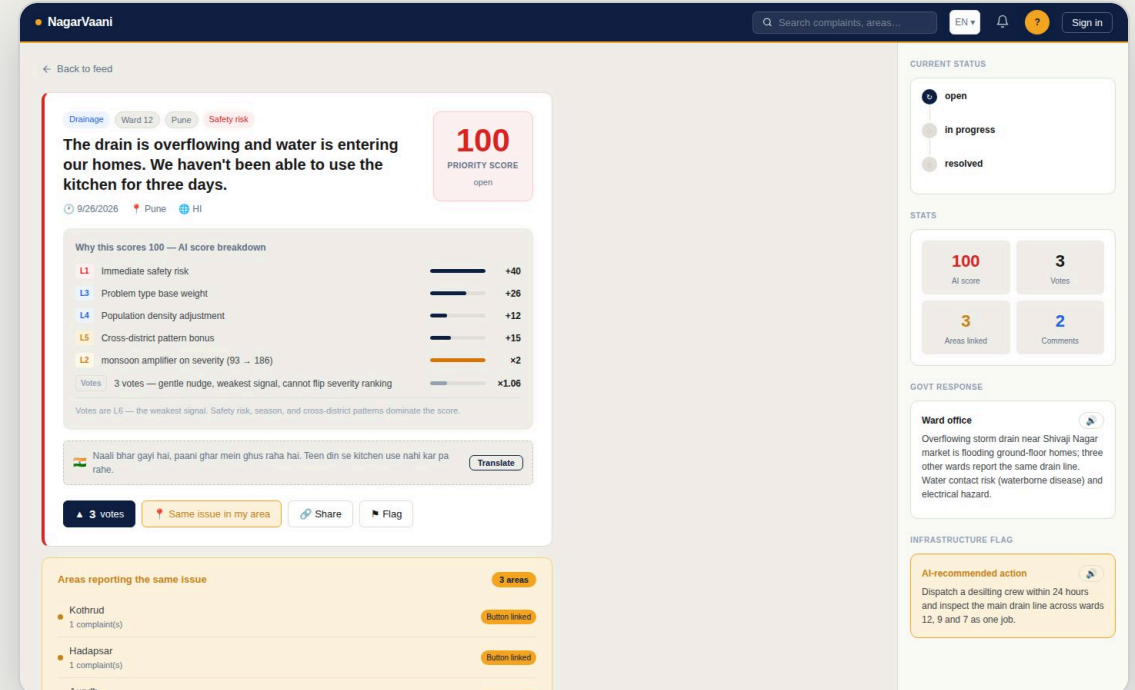
**L5 Pattern** — same issue in other areas **+5 each, max +20**

**Votes** — a gentle, bounded nudge **≤ +1 on safety**

**score** =  $\min(100, (L1 + L3 + L4 + L5) \times L2)$

safety risks → 90–99 by severity · then the vote nudge

Season is fixed when first scored — later votes can't change it.



Every complaint shows its own "Why this scores ..." breakdown — and a plain-language *How scoring works* page explains L1–L5 to citizens.

# Accountability, not triage.



## Similar issues across the whole city

Not only your ward — each with its status — so one locality's problem can be fixed together with the same problem elsewhere in the city.



## "See also across the country"

The same problem type in other cities: a pattern visible nationally, not just locally.



## Resolution counts

"In Pune: 4 reports of this kind — 1 resolved, 3 still unresolved." Public proof of what gets fixed.



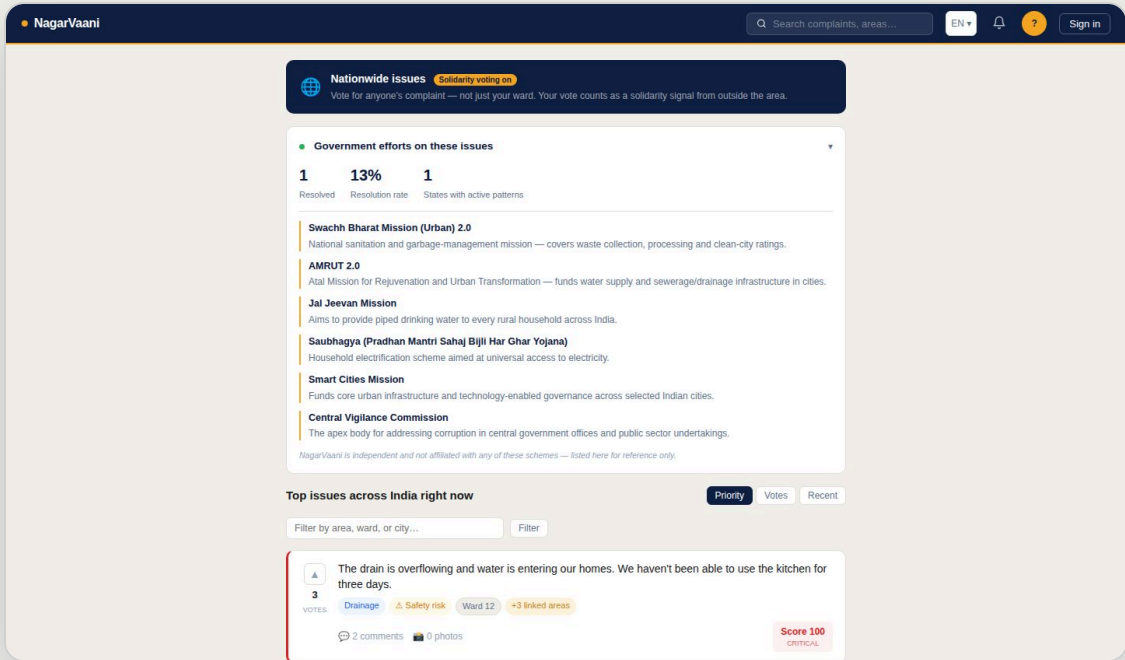
## Voters are told when it's "resolved"

...and can say if it isn't. The original reporter can **dispute** a false "resolved".

The screenshot displays the NagarVaani web application interface. At the top, there's a header with the NagarVaani logo, a search bar, and user options like 'EN', a bell icon, and a 'Sign in' button. The main content area shows a complaint titled 'Drain overflowing in Whitefield main road after every shower, - Ward 3, Bengaluru, Karnataka'. It includes a 'Score 93' badge and a 'See also across the country' link. Below the complaint, there are two comments: one from 'Ramesh Kumar' (Ward 12) dated 9/30/2026, and another from 'Ward Officer Desai' (Ward office - Official) dated 9/30/2026. The 'Ward Officer Desai' comment includes an official reply: 'Team dispatched. Desilting starts Thursday morning; we will inspect the whole line.' To the right of the main content, there's a 'CURRENT STATUS' section with a progress bar showing 'open', 'in progress', and 'resolved' stages. Below that, there's a 'STATS' section with four boxes: '100 AI score', '3 Votes', '3 Areas linked', and '2 Comments'. At the bottom right, there's a 'GOVT RESPONSE' section with a 'Ward office' response: 'Overflowing storm drain near Shivaji Nagar market is flooding ground-floor homes; three other wards report the same drain line. Water contact risk (waterborne disease) and electrical hazard.' Below that, there's an 'INFRASTRUCTURE FLAG' section with an 'AI-recommended action' box: 'Dispatch a desilting crew within 24 hours and inspect the main drain line across wards 12, 9 and 7 as one job.'

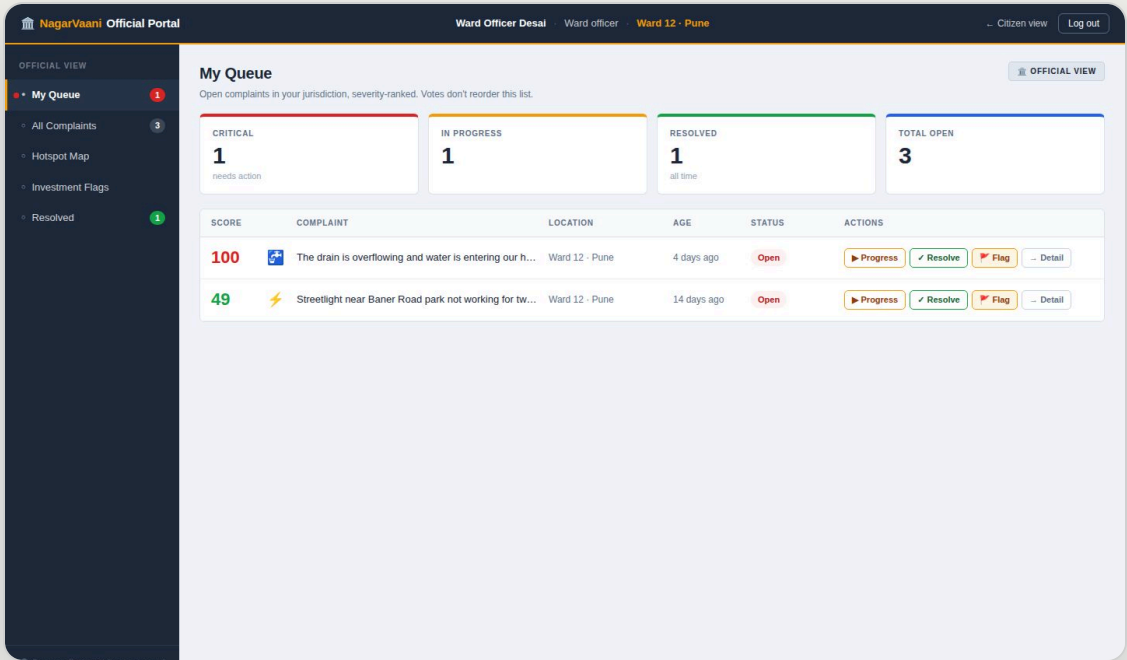
Comment thread with an official reply, automatic place detection ("Kothrud" auto-linked), translation and 🗣️ read-aloud.

# Citizens see the nation. Officials see their jurisdiction.



## Nationwide issues

Ranked by priority, with solidarity voting from outside the area. Paginated, filterable by area.



## Official's queue

Only their ward, city or state — severity-ranked. "Votes don't reorder this list."

# Where demand is highest — and which scheme funds it.

**NagarVaani Official Portal** | Nodal Officer | Central ministry | Nationwide | Citizen view | Log out

**OFFICIAL VIEW**

- My Queue (4)
- All Complaints (7)
- Hotspot Map
- Investment Flags**
- Resolved (1)
- Official Verifications

**Investment Flags**  
Complaints your office flagged as infrastructure gaps — the list a senior or state-level reviewer works from.

| COMPLAINT                                                                          | SCORE | RECOMMENDED COST | FLAGGED BY | DATE |
|------------------------------------------------------------------------------------|-------|------------------|------------|------|
| No flags yet. Use <a href="#">Flag</a> on a complaint in the queue to add it here. |       |                  |            |      |

**Data-driven project priorities**  
Unresolved demand by state and problem type, joined with Census-2011 population and mapped to the real central scheme that funds it. A triage aid — not a costed plan.

[Download CSV](#)

| State       | Problem      | Open | Avg severity | Per million people | Index        | Suggested funding scheme                            |
|-------------|--------------|------|--------------|--------------------|--------------|-----------------------------------------------------|
| Maharashtra | road         | 2    | 68.9         | 0.018              | <b>116.7</b> | PMGSY (rural) / Smart Cities Mission (urban)        |
| Maharashtra | drainage     | 1    | 99.6         | 0.009              | <b>99.6</b>  | AMRUT 2.0 (urban drainage & water)                  |
| Rajasthan   | water supply | 1    | 98.7         | 0.015              | <b>98.7</b>  | Jal Jeevan Mission / AMRUT 2.0                      |
| Maharashtra | tree hazard  | 1    | 98.4         | 0.009              | <b>98.4</b>  | Municipal horticulture / disaster-management budget |
| Karnataka   | drainage     | 1    | 93.3         | 0.016              | <b>93.3</b>  | AMRUT 2.0 (urban drainage & water)                  |
| Maharashtra | electricity  | 1    | 49.4         | 0.009              | <b>49.4</b>  | Saubhagya / state DISCOM upgrade                    |

Method: avg severity × (1 + ln(open complaints)); population = Census 2011

Scores reflect severity, season and pattern — not vote count.

## Project priorities

Unresolved demand by **state × problem type**, joined with **Census-2011 population**, mapped to the **real central scheme** (AMRUT 2.0, Jal Jeevan Mission, PMGSY...).

## Honest by design

A triage index — **not a costed plan**. We removed an early made-up ₹ figure rather than replace it with another.

## Take it away

One-click **CSV** download, live hotspot map, investment flags, admin-approved officials.

# Not another complaint box.

|                | TYPICAL PUBLIC-COMPLAINT TOOLS     | NAGARVAANI                                                                                         |
|----------------|------------------------------------|----------------------------------------------------------------------------------------------------|
| Ranking        | Votes, recency, city size          | <b>Severity × season × people × pattern</b> ; votes a small bounded signal                         |
| Language       | One language, typed                | Voice + text, <b>Hindi · Marathi · Tamil · English</b> , Hinglish, two-way translation, read-aloud |
| The AI         | A black-box score                  | <b>Explainable arithmetic</b> — AI only reads and classifies                                       |
| Follow-through | Complaint disappears into an inbox | Jurisdiction queue · status log · <b>dispute a false "resolved"</b> · voter notifications          |
| Corruption     | Mixed in with potholes             | Separate, <b>login-gated</b> channel — never in public feeds                                       |
| Openness       | Closed                             | <b>MIT licence</b> , swappable AI/speech providers, CSV export                                     |

Left column describes common patterns in complaint portals generally, not any single product.

# A working platform, not a mock-up.

## Citizens

- Report by text, mic, Telegram
- Photos, 360°, GPS → area name
- Anonymous or signed in
- Home · Trending · Near Me · Track
- Vote · comment · "same issue"
- Translate ·  read-aloud · share
- Flag abuse · dispute resolution

45

API operations · 9 routers

## Officials

- Admin-approved, work e-mail
- Ward · city · state · central
- Ranked queue & hotspot map
- Status changes inside jurisdiction
- Investment flags
- Project priorities + CSV
- Verification queue (admin)

11

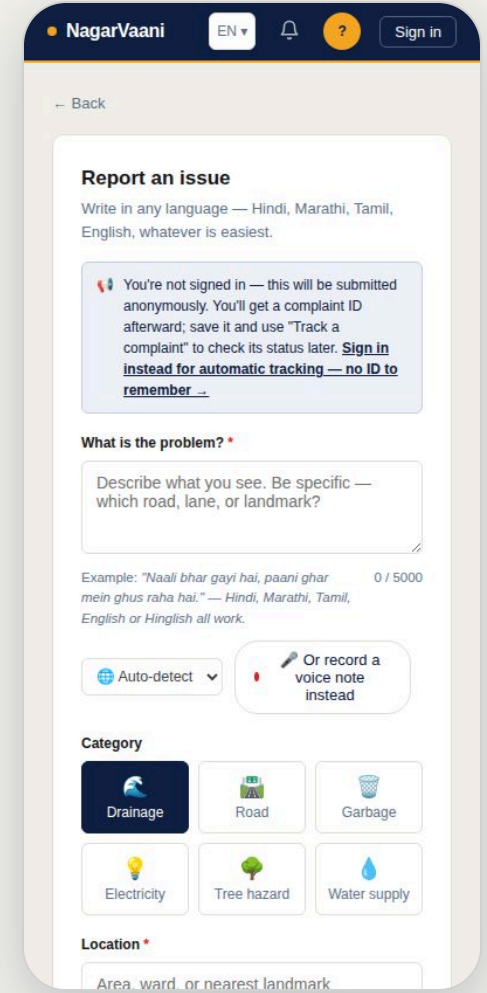
database tables

## Platform

- E-mail OTP, reset, Google sign-in
- Rate limits, security headers
- Self-registering Telegram webhook
- E-mail via Brevo + SMTP fallback
- Corruption channel, chatbot
- Neighbourhood e-mail alerts

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pages, phone-ready



The screenshot shows the NagarVaani mobile app interface. At the top, there's a header with the NagarVaani logo, a language selector (EN), a notification bell, a help icon (?), and a 'Sign in' button. Below the header, there's a 'Back' button. The main content area is titled 'Report an issue' with a subtext 'Write in any language — Hindi, Marathi, Tamil, English, whatever is easiest.' A blue box contains a message: 'You're not signed in — this will be submitted anonymously. You'll get a complaint ID afterward; save it and use "Track a complaint" to check its status later. Sign in instead for automatic tracking — no ID to remember.' Below this, there's a section 'What is the problem?' with a text input field and a placeholder 'Describe what you see. Be specific — which road, lane, or landmark?'. An example text is provided: 'Example: "Naali bhar gayi hai, paani ghar mein ghus raha hai." — Hindi, Marathi, Tamil, English or Hinglish all work.' There's a character count '0 / 5000'. Below the input field, there's a language selector 'Auto-detect' and a button 'Or record a voice note instead'. The 'Category' section has six icons: Drainage, Road, Garbage, Electricity, Tree hazard, and Water supply. The 'Location' section has a text input field with a placeholder 'Area, ward, or nearest landmark'.

# Simple, swappable, deployable on one free-tier service.

| LAYER               | CHOICE                                          |
|---------------------|-------------------------------------------------|
| Backend             | Python · FastAPI · async SQLAlchemy             |
| Database            | PostgreSQL (Neon) · SQLite locally              |
| Frontend            | Vanilla JS ES modules · Leaflet + OpenStreetMap |
| Language AI         | Gemini (default) · Claude switchable            |
| Speech in           | Whisper via Hugging Face / Groq / OpenAI        |
| Speech out          | Browser Web Speech API (no key)                 |
| Media · chat · mail | Cloudinary · Telegram Bot API · Brevo           |
| Auth                | bcrypt · JWT · e-mail OTP · Google (optional)   |
| Hosting             | Render (one service) + Neon                     |

## Architecture

 Web · Mic · Telegram →  Official portal

▼ one FastAPI service ▼

STT → Filter → Classify (LLM) → **Score (maths)** → Brief (LLM)

Translation · place detection · chatbot · alerts



PostgreSQL · Cloudinary · Brevo · Telegram

No framework, no build step — the frontend is served by the same process as the API.

# What's real, what's a stub, what's demo.

## REAL — proper backend

- Intake: form, voice, Telegram
- AI pipeline & L1–L5 scoring
- Votes, comments, cross-area links
- Status workflow + **jurisdiction enforcement**
- Disputes & e-mail/Telegram notifications
- Auth, OTP, admin-approved officials
- Hotspot map & statistics from the DB
- Priorities + CSV · translation · read-aloud
- Rate limits & security headers

## STUB — provision built

- Population factor = default 10,000  
(`District` table not wired)
- **No** infrastructure-index / investment data
- **No** project costs — none invented
- Flag moderation: API only, no screen
- Volunteer forms → e-mail only
- "Our Impact" tiles: coming soon
- Schemes page: fixed list

## DEMO — labelled

- Seed users & sample complaints
- "Supporters" placeholder cards
- Example cards before data loads

## NOT BUILT

- WhatsApp / SMS
- Other countries
- Offline / installable

# We attacked our own system — and fixed what we found.

76

scripted endpoint checks — auth, roles, jurisdiction, limits, bad input — all passing

16

pages verified in a real browser at desktop and phone width — zero JS errors

1

critical account-takeover path found and fixed before submission

| SEVERITY | FINDING                                                                           | FIX                                                                        |
|----------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Critical | Verification endpoint logged in any verified account; API exposed the author's id | Refuses verified accounts; only an <code>is_author</code> flag is returned |
| High     | No brute-force limits on login / 6-digit OTP                                      | Per-account & per-IP limits (429)                                          |
| High     | An official could change <i>any</i> complaint's status                            | Enforced inside their jurisdiction                                         |
| Medium   | Forgeable official flag; <code>javascript:</code> image URLs                      | Reserved prefix; https-only, no inline JS                                  |
| Medium   | Anonymous endpoints could burn AI credit                                          | Per-IP limits, input caps, security headers                                |

Still on the list: token in `localStorage` , in-memory rate limiter, no CSP yet, AI filter fails open.

# Strong on intake and prioritisation. Honest about the data gaps.

| REQUIREMENT                               | STATUS  | WHERE WE ARE                                                                        |
|-------------------------------------------|---------|-------------------------------------------------------------------------------------|
| Voice, text, messaging intake             | Done    | Web · mic · Telegram (WhatsApp not built)                                           |
| Multilingual                              | Done    | 4 UI languages · AI understands more · two-way translation                          |
| Aggregate & surface hotspots              | Done    | Scoring, cross-area patterns, live map, ranked priorities                           |
| Combine with demographic data             | Partial | Census-2011 state populations in priorities; per-area density still a constant      |
| Infrastructure indices · investment plans | Gap     | Not ingested — the largest gap                                                      |
| Recommend priority projects               | Partial | Transparent triage index + matching central scheme; no costs                        |
| Measure impact                            | Partial | Status log, disputes, resolved counts; no outcome dashboard yet                     |
| Digital Public Good                       | Partial | MIT licence · privacy note · swappable providers; open-data export pending          |
| BRICS                                     | Gap     | India only — scoring & dashboards are country-agnostic; the <b>data pack</b> is not |

# Every gap is data or a channel — not a rewrite.

## Now — done

- Intake → scoring → officials
- Hotspots, priorities + CSV
- Translation, read-aloud
- Security hardening, deploy

## Next — weeks

- Real district population + one infrastructure dataset → **investment-gap** view
- Impact dashboard: time-to-resolve, dispute rate, **neglect index**
- Moderation screen; duplicate merging
- Installable / offline web app

## Later — months

- **BRICS:** `country` field + data pack (states, schemes, languages) per nation
- WhatsApp & SMS/IVR channels
- Costed recommendations from schedules of rates
- Open-data export → full Digital Public Good

**Hardening on the way:** Redis rate limiting · Content-Security-Policy · httpOnly cookies · human-review queue when AI is unavailable.

## HOW IT WAS BUILT

# One developer. Real decisions. AI as a tool.

NagarVaani was **conceived, specified and driven by one developer** — the problem framing, the scoring philosophy, what to build and what to leave out, deployment and hands-on testing of every flow.

AI coding assistants wrote and reviewed code faster, the way a compiler or linter helps — while the developer set the requirements, rejected what didn't fit, and verified the result.

The judgement shows: we **removed an invented ₹ figure** instead of replacing it, and **fixed a critical flaw** found in our own security review.

## TRY IT

**<https://nagarvaani-2i2l.onrender.com>**

Code & docs: [github.com/FalguniTakawale/NagarVaani](https://github.com/FalguniTakawale/NagarVaani)

Telegram: [@NagarvaaniHackBot](https://t.me/NagarvaaniHackBot)

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**Low votes ≠ low importance.**